

Problem Set 4: Solutions

SVARs: Short-Run and Long-Run Identification, and VECMs

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Problem Set 4 Solutions

- 1. Introduction: The Identification Problem in Macroeconometrics**
 - ▶ SVMA representation, fundamental non-identification, rotation parameterization
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 - ▶ Long-run productivity restriction, CEE, identification analysis, computational procedure
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 - ▶ Granger representation, structural long-run impact, noninvertibility of Δx_t , SVECM vs VAR in levels
- 4. Problem 3: CFP Critique — Cholesky Under Misspecification**
 - ▶ Identified shocks as mixtures, sufficient condition for sign reversal, robustness

Introduction

The Identification Problem in Macroeconometrics

Objects of interest. Most Macroeconomic Models can be expressed under SVMA representation:

$$y_t = \sum_{\ell=0}^{\infty} \Theta_{\ell} \varepsilon_{t-\ell}, \quad \varepsilon_t \sim WN(0, I_n).$$

- ▶ From $\{\Theta_{\ell}\}$ we get: **IRFs, FVDs, Historical Decompositions**
- ▶ These are the key objects for policy analysis and testing macro models

How can we learn $\{\Theta_{\ell}\}$ from data? Three broad approaches:

1. **Semi-structural, time series:** use time-series properties of y_t + identifying assumptions for *families* of models ← our focus today
2. **Semi-structural, cross-sectional:** microeconomic causal designs
3. **Full structural:** specify and estimate a DSGE model

The Fundamental Identification Problem

Proposition (Non-identification from second moments)

Under standard assumptions, we can find $\{\Theta_\ell^*\}$ and $\{\varepsilon_t^*\}$ such that $\Theta_\ell \neq \Theta_\ell^*$ for some ℓ , $\varepsilon_t^* \sim WN(0, I_n)$, and for all h :

$$\Gamma_h = \text{Cov} \left(\sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell}, \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell-h} \right) = \sum_{\ell=0}^{\infty} \Theta_\ell^* (\Theta_{\ell+h}^*)'.$$

Proof sketch. Take any orthogonal Q ($QQ' = I_n$, $Q \neq I$). Set $\Theta_\ell^* = \Theta_\ell Q$ and $\varepsilon_t^* = Q' \varepsilon_t$. Then:

$$\sum_{\ell} \Theta_\ell \Theta_{\ell+h}' = \sum_{\ell} \Theta_\ell Q Q' \Theta_{\ell+h}' = \sum_{\ell} (\Theta_\ell Q) (\Theta_{\ell+h} Q)' = \sum_{\ell} \Theta_\ell^* (\Theta_{\ell+h}^*)'.$$

$\Rightarrow$ We can always rotate the shocks and preserve the entire serial correlation structure.

Shocks vs. Prediction Errors

The Wold representation (reducible-form):

$$y_t = \mu + \sum_{\ell=0}^{\infty} B_{\ell} u_{t-\ell}, \quad u_t \sim WN(0, \Sigma_u).$$

Under invertibility (shocks are linear combinations of present and past data):

$$u_t = H\varepsilon_t \implies \Sigma_u = HH', \quad B_{\ell} = \Theta_{\ell}H^{-1}.$$

- ▶ We can estimate $\{B_{\ell}\}$ and Σ_u from data
- ▶ **But** this is not enough to recover $\{\Theta_{\ell}\}$

The rotation parameterization. Write $H = \text{Ch}(\Sigma_u) \cdot Q$ where $\text{Ch}(\Sigma_u)$ is the Cholesky square root:

$$\Sigma_u = HH' \iff H = \text{Ch}(\Sigma_u) Q, \quad QQ' = I_n.$$

Task: impose restrictions on $\{\Theta_{\ell}\}$ to pin down a unique Q .

Order condition. H has n^2 parameters but $\Sigma_u = HH'$ provides only $\frac{n(n+1)}{2}$ restrictions $\Rightarrow$ need $\frac{n(n-1)}{2}$ additional restrictions.

Two broad families:

1. **Invertibility + restrictions** (classical VAR literature):
 - ▶ Zero (short-run or long-run), sign, statistical
2. **Without invertibility** (recent):
 - ▶ Proxy/Narrative SVARs, macro IVs

This problem set covers:

- ▶ **Problem 1:** Long-run zeros (Galí 1999) + short-run zeros (CEE 2005)
- ▶ **Problem 2:** Long-run restrictions in cointegrated systems (SVECM)
- ▶ **Problem 3:** Critique — what goes wrong with Cholesky (CFP 2009)

Problem 1: Hybrid Identification

Galí (1999) Long-Run + CEE (2005) Short-Run Restriction

Problem 1: Setup

VAR and Structural Form

$$y_t = (\Delta a_t, n_t, \pi_t, i_t)', \quad y_t = \sum_{j=1}^p A_j y_{t-j} + u_t, \quad \mathbb{E}[u_t u_t'] = \Sigma_u.$$

Structural form: $u_t = B_0^{-1} w_t$, $\mathbb{E}[w_t w_t'] = I_4$, $w_t = (\varepsilon_t^a, \varepsilon_t^d, \varepsilon_t^\pi, \varepsilon_t^m)'$.

Structural MA: $y_t = \sum_{h=0}^{\infty} \Theta_h w_{t-h}$, $\Theta_0 = B_0^{-1}$.

Long-run multiplier: $\Theta(1) = \sum_{h=0}^{\infty} \Theta_h = \Phi(1) B_0^{-1}$.

Variables:

- ▶ Δa_t : productivity growth n_t : hours worked
- ▶ π_t : inflation i_t : interest rate
- ▶ ε_t^a : technology ε_t^d : demand ε_t^π : cost-push ε_t^m : monetary

Problem 1(a): Galí's Long-Run Restriction — Derivation

Galí's assumption: Only the technology shock ε_t^a has a permanent effect on the *level* of productivity a_t .

Mapping VAR growth rates to levels. Since $a_T = a_{-1} + \sum_{t=0}^T \Delta a_t$, substitute the structural MA for Δa_t :

$$\frac{\partial a_T}{\partial \varepsilon_0^j} = \sum_{t=0}^T \sum_{h=0}^{\infty} [\Theta_h]_{1,j} \mathbf{1}\{t-h=0\} = \sum_{t=0}^T [\Theta_t]_{1,j}.$$

Taking $T \rightarrow \infty$:

$$\lim_{T \rightarrow \infty} \frac{\partial a_T}{\partial \varepsilon_0^j} = \sum_{t=0}^{\infty} [\Theta_t]_{1,j} = [\Theta(1)]_{1,j}.$$

The long-run effect on productivity *levels* equals the $(1, j)$ entry of $\Theta(1)$.

Problem 1(a): Galí's Long-Run Restriction — Result

Identifying restriction

Only ε_t^a ($j = 1$) has a long-run effect on productivity levels. For $j \in \{2, 3, 4\}$:

$$[\Theta(1)]_{1,2} = [\Theta(1)]_{1,3} = [\Theta(1)]_{1,4} = 0.$$

Equivalently: $[\Theta(1)]_{1,\cdot} = (* \quad 0 \quad 0 \quad 0)$.

Connection to reduced form. Since $\Theta(1) = \Phi(1) B_0^{-1} = A(1)^{-1} B_0^{-1}$:

$$[\Theta(1)]_{1,j} = 0 \iff \text{first row of } A(1)^{-1} B_0^{-1} \text{ has zeros in cols 2,3,4.}$$

Three long-run restrictions on the product $A(1)^{-1} B_0^{-1}$.

Problem 1(b): CEE Short-Run Restriction

Christiano–Eichenbaum–Evans (2005): Productivity growth does not respond contemporaneously to a monetary policy shock:

$$\left. \frac{\partial(\Delta a_t)}{\partial \varepsilon_t^m} \right|_t = 0.$$

The contemporaneous response equals the impact matrix entry:

$$\left. \frac{\partial(\Delta a_t)}{\partial \varepsilon_t^m} \right|_t = [B_0^{-1}]_{1,4} = 0.$$

Short-run restriction

$[B_0^{-1}]_{1,4} = 0$: the (1, 4) entry of the impact matrix is zero.

Interpretation: Within the period, a monetary shock cannot affect productivity growth (no contemporaneous supply-side channel).

Problem 1(c): Identification Analysis

Degrees of freedom. B_0^{-1} is 4×4 (16 elements). Normalization $B_0^{-1}(B_0^{-1})' = \Sigma_u$ gives $\frac{4 \times 5}{2} = 10$ restrictions. **Remaining d.o.f.:** $16 - 10 = 6$.

Restrictions available:

- ▶ 3 long-run restrictions from (a): $[\Theta(1)]_{1,2} = [\Theta(1)]_{1,3} = [\Theta(1)]_{1,4} = 0$
- ▶ 1 short-run restriction from (b): $[B_0^{-1}]_{1,4} = 0$
- ▶ **Total: 4 restrictions < 6 d.o.f. $\Rightarrow$ under-identified by the order condition**

Rank condition. Write $B(\theta) = \hat{L}Q(\theta)$ ($\hat{L}$ = Cholesky of $\hat{\Sigma}_u$, $Q(\theta)$ orthogonal). Stack restrictions as $f(\theta) = 0$:

$$f(\theta) = \begin{pmatrix} (\hat{L}Q)_{1,4} \\ (\hat{\Phi}(1)\hat{L}Q)_{1,2} \\ (\hat{\Phi}(1)\hat{L}Q)_{1,3} \\ (\hat{\Phi}(1)\hat{L}Q)_{1,4} \end{pmatrix} = 0.$$

Local identification requires $\text{rank}(\partial f / \partial \theta') = 4$ at the true θ_0 . **2 additional restrictions needed** for exact identification.

Problem 1(d): Computational Procedure

Core idea: estimate reduced form by OLS, then search over rotations $Q(\theta)$ until $f(\theta) = 0$.

1. **Estimate VAR by OLS.** Obtain $(\hat{A}_1, \dots, \hat{A}_p)$, residuals $\hat{u}_t$, and $\hat{\Sigma}_u$.
2. **Compute reduced-form long-run multiplier.**

$$\hat{A}(1) = I - \hat{A}_1 - \dots - \hat{A}_p, \quad \hat{\Phi}(1) = \hat{A}(1)^{-1}.$$

3. **Factor $\hat{\Sigma}_u$ and parameterize Q .** Compute Cholesky $\hat{L}$ ($\hat{\Sigma}_u = \hat{L}\hat{L}'$).
Parameterize $Q(\theta)$ as Givens rotations
4. **Solve $f(\theta) = 0$.** With 2 additional restrictions: square 6×6 system, use root-finder. Without: set-identified — report range.
5. **Recover structural objects.** $\hat{B}_0^{-1} = \hat{L}Q(\hat{\theta})$, $\hat{\Theta}_h = \hat{\Phi}_h\hat{B}_0^{-1}$.

Problem 1(e): Hours Fall After a Technology Shock

Galí's finding: hours worked n_t *fall* after a positive technology shock — contrary to standard RBC predictions.

Identified pattern:

- ▶ $[\Theta_0]_{2,1} < 0$: hours fall on impact
- ▶ Hours may remain below steady state for several periods
- ▶ Long run: hours return to (or slightly above) steady state
- ▶ Inflation: $[\Theta_h]_{3,1} < 0$ for small h — technology reduces marginal cost

Reconciliation with long-run restriction:

- ▶ $[\Theta(1)]_{1,1} > 0$: technology permanently raises productivity (✓)
- ▶ Mechanism: sticky prices $\Rightarrow$ firms cannot lower prices immediately $\Rightarrow$ reduce labor input
- ▶ In the long run, prices adjust, output expands, productivity remains permanently higher

The long-run restriction on productivity does *not* constrain the short-run response of hours.

Problem 2: SVECM

Permanent and Transitory Shocks in a Cointegrated System

Definition

x_t ($I(1)$, $n \times 1$) is **cointegrated** if $\exists$ nonzero β s.t. $\beta'x_t \sim I(0)$.

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- ▶ Some shocks are **permanently non-neutral** (technology); others are **neutral in the long run** (monetary)
- ▶ Cointegration rank $r \Rightarrow$ exactly $n - r$ **common stochastic trends** $\Rightarrow$ at most $n - r$ permanent shocks
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We want to find a representation which is stationary and uses cointegration structure!

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Fundamental problem (Beveridge–Nelson)

If $x_t \sim I(1)$, then Δx_t has no VAR representation (despite stationary)

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If we correct for non-invertibility, we get a VAR-like representation: the **VECM**.

VECM (follows from Granger Representation Theorem)

If $x_t \sim I(1)$ with cointegration rank r , then Δx_t is stationary and can be represented as:

$$\Delta x_t = \underbrace{\alpha \beta' x_{t-1}}_{\text{error correction}} + \sum_{j=1}^{p-1} \Gamma_j \Delta x_{t-j} + u_t$$

where β : cointegrating vectors; and α : adjustment (loading) coefficients

Background: From VECM to SVECM

Add structural layer $u_t = B_0^{-1}w_t$, $\mathbb{E}[w_t w_t'] = I_n$ — exactly as in a standard SVAR.

What cointegration buys for identification:

- ▶ Non-stationary long-run level representation:

$$x_t = C \sum_{s=1}^t u_s + C^*(L)u_t + x_0^*,$$

where C is the **reduced-form long-run multiplier** (cumulated impact on levels)

- ▶ **Structural long-run impact:** since $u_t = B_0^{-1}w_t$, the long-run impact of w_t on x_t is $\Upsilon = C B_0^{-1}$
- ▶ $\text{rank}(C) = n - r \Rightarrow$ only $n - r$ **permanent shocks**; transitory shocks have **zero columns in Υ**

This is the Shapiro–Watson (1988) / KPSW (1991) approach — exactly what Problem 2 asks you to implement.

Problem 2: Setup

VECM for $x_t = (y_t, c_t, i_t)'$

$$\Delta x_t = \alpha \beta' x_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta x_{t-j} + u_t, \quad \mathbb{E}[u_t u_t'] = \Sigma_u.$$

x_t is $I(1)$, cointegration rank $r = 1$. $\alpha, \beta \in \mathbb{R}^{3 \times 1}$.

Structural shocks: $u_t = B_0^{-1} w_t$, $w_t = (\varepsilon_t^P, \varepsilon_t^T, \varepsilon_t^M)'$, $\mathbb{E}[w_t w_t'] = I_3$.

The VECM implies (**Granger Representation Theorem**):

$$x_t = C \sum_{s=1}^t u_s + C^*(L) u_t + x_0^*, \quad C = \beta_{\perp} (\alpha'_{\perp} \Gamma(1) \beta_{\perp})^{-1} \alpha'_{\perp}.$$

- ▶ Orthogonal Complement: $\beta' \beta_{\perp} = 0$
- ▶ $C \sum u_s$: stochastic trends (permanent component)
- ▶ $C^*(L) u_t$: transitory dynamics
- ▶ $\text{rank}(C) = n - r = 2$: exactly 2 common stochastic trends

Problem 2(a): Common Stochastic Trends

Result

With $n = 3$ variables and cointegration rank $r = 1$:

$$\text{Number of common stochastic trends} = n - r = 3 - 1 = 2.$$

Implication for structural identification:

- ▶ Since $\text{rank}(C) = 2$, at most **2 shocks can have permanent effects** on the levels
- ▶ The third shock must be **transitory** (no long-run effect on any variable)
- ▶ This is **not an identifying assumption** — it is a consequence of cointegration
- ▶ It provides 1 free restriction; we need 1 more for exact identification

Problem 2(b)–(c): Long-Run Structural Restrictions

Structural long-run impact matrix. Substituting $u_t = B_0^{-1}w_t$ into the Granger representation:

$$x_t = \underbrace{CB_0^{-1} \sum_{s=1}^t w_s}_{\text{permanent}} + \underbrace{C^*(L)B_0^{-1}w_t}_{\text{transitory}} + x_0^*.$$

$\Upsilon \equiv CB_0^{-1}$: structural long-run impact matrix on levels.

(b) Transitory shock ε_t^T (Shapiro–Watson / KPSW)

$[\Upsilon]_{\cdot,2} = \mathbf{0}$: no long-run effect on output, consumption, or investment. *Note:* $\text{rank}(C) = 2$ makes only 2 of these 3 restrictions independent.

(c) Monetary neutrality (long run)

$[\Upsilon]_{1,3} = 0$: monetary/financial shock has no long-run effect on output.

Problem 2(d): Short-Run Restriction and Identification

(d) Short-run restriction (CEE-style)

Monetary shock does not affect output growth contemporaneously:

$$[B_0^{-1}]_{1,3} = 0.$$

Identification count:

- ▶ B_0^{-1} is 3×3 : $9 - 6 = 3$ d.o.f.
- ▶ Restrictions: 2 independent (from (b)) + 1 (from (c)) + 1 (from (d)) = 4
- ▶ **Over-identified by the count** — need to verify consistency

Key feasibility check. Since $\text{rank}(C) = 2$, column 2 of Υ is zero iff $[B_0^{-1}]_{\cdot,2}$ lies in $\ker(C)$ (1-dimensional). This restricts column 2 of B_0^{-1} to a line — numerical verification required for specific $(\hat{C}, \hat{\Sigma}_u)$.

Problem 2(f): Why a VAR in First Differences Is Invalid

Wold representation of Δx_t :

$$\Delta x_t = \delta + \psi(L)\varepsilon_t, \quad \psi(1) = \text{long-run multiplier matrix.}$$

Key result (Beveridge–Nelson / Granger)

Cointegration ($r = 1$) $\Rightarrow \text{rank}(\psi(1)) = n - r = 2 < 3 \Rightarrow |\psi(1)| = 0$.

$L = 1$ is a root of $|\psi(L)| = 0 \Rightarrow$ unit MA root $\Rightarrow \psi(L)$ is **noninvertible**.

Δx_t does not have a VAR representation.

A finite-order VAR in first differences is fundamentally invalid for a cointegrated system—one cannot “flip” a unit root on the unit circle.

Problem 2(f): SVECM vs. VAR in Levels

The correct comparison: SVECM vs. unrestricted VAR in levels
($x_t = \sum_j A_j x_{t-j} + u_t$).

SVECM

- + Exploits reduced-rank structure $\alpha\beta' \Rightarrow$ more efficient
- **Requires correct specification of r**

VAR in levels

- + Consistent regardless of true r (Stock, 1987); no rank commitment needed
- Less efficient; non-standard inference on individual coefficients

Failure modes (SVECM):

- ▶ **r overspecified** ($r = 2$, true $r = 1$): $\text{rank}(C)$ too low $\Rightarrow \Upsilon$ inconsistent
- ▶ **r underspecified** ($r = 0$): reduces to VAR in $\Delta x_t \Rightarrow$ invalid (noninvertible MA)

Problem 3: The CFP Critique

Recursive Identification Under Misspecification

Problem 3: Setup

True DGP and Econometrician's Model

True DGP: $z_t = \sum_{h=0}^{\infty} \Psi_h \varepsilon_{t-h}$, $\varepsilon_t = (\varepsilon_t^a, \varepsilon_t^c, \varepsilon_t^m)'$, $\mathbb{E}[\varepsilon_t \varepsilon_t'] = I_3$.

Econometrician: estimates VAR, imposes Cholesky ordering (y, π, i) .

$u_t = B_0^{-1} w_t$ with B_0^{-1} **lower triangular**.

Key assumption: In the true DGP, monetary policy affects output *contemporaneously*:

$$(\Psi_0)_{1,3} \neq 0 \quad (\text{true DGP}).$$

But Cholesky imposes:

$$(B_0^{-1})_{1,3} = 0 \quad (\text{identification assumption}).$$

These two conditions are contradictory — this is the source of misidentification.

Problem 3(a): Cholesky Zero Restrictions

With ordering (y, π, i) , the lower-triangular B_0^{-1} imposes:

$$B_0^{-1} = \begin{pmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{pmatrix}.$$

Which variables respond to which shocks (contemporaneously):

	w_t^1	w_t^2	w_t^3 (“monetary”)
y_t	✓	×	×
π_t	✓	✓	×
i_t	✓	✓	✓

Economic content: the monetary shock affects only the interest rate on impact — neither output nor inflation respond within the period.

Problem 3(b): Non-Uniqueness and Cholesky

The non-uniqueness problem. $\Sigma_u = B_0^{-1}(B_0^{-1})'$ has $\frac{3 \cdot 4}{2} = 6$ equations in 9 unknowns $\Rightarrow$ 3 d.o.f. remain. For any orthogonal Q :

$$\tilde{B}_0^{-1} = B_0^{-1}Q \implies \tilde{B}_0^{-1}(\tilde{B}_0^{-1})' = B_0^{-1}QQ'(B_0^{-1})' = \Sigma_u.$$

Infinitely many valid impact matrices.

How Cholesky selects a rotation. Let $\Sigma_u^{1/2}$ be the lower-triangular Cholesky square root. Cholesky sets:

$$B_0^{-1} = \Sigma_u^{1/2} \iff Q = I \quad \text{in the representation } B_0^{-1} = \Sigma_u^{1/2}Q.$$

- ▶ This is an **economic assumption about timing**, not a statistical necessity
- ▶ Different variable orderings yield different identified shocks
- ▶ Assuming $Q = I$ is valid *only if* $B_0^{-1} = \Psi_0$ (correct identification)

Problem 3(c): Identified Shock as a Mixture — Derivation

Step 1: Reduced-form innovation $= \Psi_0 \varepsilon_t$. From the structural MA $z_t = \sum_{h \geq 0} \Psi_h \varepsilon_{t-h}$, under invertibility:

$$u_t = z_t - \text{Proj}(z_t \mid \mathcal{F}_{t-1}^z) = \sum_{h \geq 0} \Psi_h \varepsilon_{t-h} - \sum_{h \geq 1} \Psi_h \varepsilon_{t-h} = \Psi_0 \varepsilon_t.$$

Step 2: Econometrician applies Cholesky. $w_t = B_0 u_t = B_0 \Psi_0 \varepsilon_t$. Define:

$$R \equiv B_0 \Psi_0, \quad w_t = R \varepsilon_t.$$

Under correct identification: $B_0^{-1} = \Psi_0 \Rightarrow R = I$. **Here $R \neq I$.**

Step 3: Proof that w_t^m is a mixture. If $R_{31} = R_{32} = 0$, then $[B_0^{-1}]_{\cdot,3} = [\Psi_0]_{\cdot,3}$. But $[B_0^{-1}]_{1,3} = 0$ (Cholesky) and $[\Psi_0]_{1,3} \neq 0$ (true DGP) — **contradiction**.

Result

$w_t^m = R_{31} \varepsilon_t^a + R_{32} \varepsilon_t^c + R_{33} \varepsilon_t^m$, with $(R_{31}, R_{32}) \neq (0, 0)$.

Problem 3(d): Sufficient Condition for Sign Reversal — Step 1

IRF linearity. By linearity of impulse responses:

$$\text{IRF}_y^{w^m}(h) = R_{31} \cdot \text{IRF}_y^{\varepsilon^a}(h) + R_{32} \cdot \text{IRF}_y^{\varepsilon^c}(h) + R_{33} \cdot \text{IRF}_y^{\varepsilon^m}(h).$$

Evaluate at $h = 0$. Since $\text{IRF}_y^{\varepsilon^j}(0) = (\Psi_0)_{1,j}$:

$$\text{IRF}_y^{w^m}(0) = R_{31}(\Psi_0)_{1,1} + R_{32}(\Psi_0)_{1,2} + R_{33}(\Psi_0)_{1,3}.$$

But Cholesky forces $(B_0^{-1})_{1,3} = 0$, so $\text{IRF}_y^{w^m}(0) = 0$. Therefore:

$$R_{33}(\Psi_0)_{1,3} = -[R_{31}(\Psi_0)_{1,1} + R_{32}(\Psi_0)_{1,2}]. \quad (*)$$

Remarks

- ▶ (*) is **automatic** — it holds for any Ψ_0 , not an extra assumption
- ▶ It shows: at $h = 0$, the monetary term and contamination term are *equal and opposite*
- ▶ Sign reversal at $h = 0$ is **impossible by construction**; it manifests at $h \geq 1$

Problem 3(d): Sufficient Condition for Sign Reversal — Step 2

Sufficient condition (two parts)

1. Restriction on Ψ_0 (contemporaneous impact matrix):

$$(\Psi_0)_{1,3} < 0 \quad (\text{true monetary shock lowers output on impact}).$$

Since $R_{33} > 0$, constraint (*) implies $R_{31}(\Psi_0)_{1,1} + R_{32}(\Psi_0)_{1,2} > 0$: the contamination enters with the *opposite* sign, hidden at $h = 0$ by exact cancellation.

2. Dynamic restriction: $\exists h \geq 1$ such that $(\Psi_h)_{1,3} < 0$ and:

$$R_{31}(\Psi_h)_{1,1} + R_{32}(\Psi_h)_{1,2} > |R_{33}(\Psi_h)_{1,3}|.$$

Conclusion. Under these conditions, $\text{IRF}_y^{w^m}(h) > 0$ while $\text{IRF}_y^{\varepsilon^m}(h) < 0$: the identified IRF has the wrong sign.

Problem 3(e): Two Robustness Strategies

Strategy 1: Sign Restrictions (Uhlig 2005)

- ▶ Replace exact zeros with *inequality* constraints on IRFs:
 - ▶ $\text{IRF}_i^{\varepsilon^m}(0) > 0$, $\text{IRF}_y^{\varepsilon^m}(h) < 0$, $\text{IRF}_\pi^{\varepsilon^m}(h) < 0$
- ▶ **Identifies:** a *set* of valid rotations $\Rightarrow$ set-identified IRFs
- ▶ **Advantage:** robust to timing misspecification

Strategy 2: Proxy SVAR / External Instruments (Gertler–Karadi 2015, Romer–Romer 2004)

- ▶ Use instrument z_t correlated with ε_t^m and uncorrelated with $(\varepsilon_t^a, \varepsilon_t^c)$
- ▶ IRF identified (up to scale) via IV:

$$\text{IRF}_y^{\varepsilon^m}(h) \propto \frac{\text{Cov}(y_{t+h}, z_t)}{\text{Cov}(i_t, z_t)}.$$

- ▶ **Advantage:** no timing assumptions; identification from external information

Big picture:

- ▶ All identification strategies are about pinning down the rotation Q in $H = \text{Ch}(\Sigma_u)Q$
- ▶ Short-run and long-run restrictions are two complementary tools
- ▶ Misspecified restrictions (like Cholesky when $(\Psi_0)_{1,3} \neq 0$) can flip the sign of IRFs
- ▶ Always check economic plausibility of identifying assumptions